

CS306 – Database Systems

Term Project

Phase 2

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Project Title:

Gym Management System

Relational Schema

```
CREATE TABLE Branch(  
    bid INT PRIMARY KEY,  
    bname VARCHAR(100) NOT NULL,  
    bcity VARCHAR(50),  
    bphone VARCHAR(20)  
)
```

```
CREATE TABLE Employee(  
    ssn CHAR(11) PRIMARY KEY,  
    ename VARCHAR(50) NOT NULL,  
    ephone VARCHAR(20),  
    job_title VARCHAR(30),  
    salary DECIMAL(10,2),  
    bid INT NOT NULL,  
    FOREIGN KEY (bid) REFERENCES Branch(bid) ON DELETE CASCADE  
)
```

```
CREATE TABLE Equipment(  
    eqid INT PRIMARY KEY,  
    `type` VARCHAR(50),  
    `condition` VARCHAR(50),  
    purchase_date DATE,  
    bid INT NOT NULL,  
    FOREIGN KEY (bid) REFERENCES Branch(bid) ON DELETE CASCADE)
```

```
CREATE TABLE Member(
  mid INT PRIMARY KEY,
  mname VARCHAR(50) NOT NULL,
  mage INT,
  maddress VARCHAR(255)
)
```

```
CREATE TABLE Class(
  cid INT PRIMARY KEY,
  cname VARCHAR(100) NOT NULL,
  capacity INT,
  duration INT,
  ssn CHAR(11) NOT NULL,
  FOREIGN KEY (ssn) REFERENCES Employee(ssn) ON DELETE CASCADE
)
```

```
CREATE TABLE Emergency_Contact (
  mid INT,
  ecname VARCHAR(50),
  ecphone VARCHAR(20),
  relation VARCHAR(50),
  PRIMARY KEY (mid, ecname),
  FOREIGN KEY (mid) REFERENCES Member(mid) ON DELETE CASCADE
)
```

```
CREATE TABLE Enrolled(
  mid INT,
  cid INT,
  enrollment_date DATE,
  PRIMARY KEY(mid,cid),
  FOREIGN KEY (mid) REFERENCES Member(mid) ON DELETE CASCADE,
  FOREIGN KEY (cid) REFERENCES Class(cid) ON DELETE CASCADE
)
```

Mapping Explanation

The ER diagram was converted into relational tables using standard mapping rules.

- **Entities** were mapped to tables with their attributes. Each entity has a primary key (e.g., Branch(bid), Member(mid)).
- **One-to-Many (1:N) relationships** were handled by adding a foreign key to the table on the many side.
For example, Employee and Equipment include bid as a foreign key referencing Branch.
- **Many-to-Many (M:N) relationships** were converted into a separate relation.
The Enrolled table was created to represent the relationship between Member and Class, with a composite primary key (mid, cid).

- **Weak entities** were mapped by creating a table with a composite primary key. Emergency_Contact depends on Member and uses (mid, ecname) as its primary key.
- **Constraints** such as primary keys, foreign keys, and NOT NULL constraints were applied to ensure data integrity.

Action Output				
#	Time	Action	Message	Duration / Fetch
✓ 1	16:53:13	CREATE DATABASE gym_app	1 row(s) affected	0.031 sec
✓ 2	16:53:13	USE gym_app	0 row(s) affected	0.000 sec
✓ 3	16:53:13	CREATE TABLE Branch(bid INT PRIMARY KEY, bna...	0 row(s) affected	0.016 sec
✓ 4	16:53:13	CREATE TABLE Employee(ssn CHAR(11) PRIMARY KE...	0 row(s) affected	0.031 sec
✓ 5	16:53:13	CREATE TABLE Equipment(eqid INT PRIMARY KEY, ...	0 row(s) affected	0.031 sec
✓ 6	16:53:13	CREATE TABLE Member(mid INT PRIMARY KEY, mn...	0 row(s) affected	0.016 sec
✓ 7	16:53:13	CREATE TABLE Class(cid INT PRIMARY KEY, cnam...	0 row(s) affected	0.031 sec
✓ 8	16:53:13	CREATE TABLE Emergency_Contact (mid INT, ecna...	0 row(s) affected	0.032 sec
✓ 9	16:53:13	CREATE TABLE Enrolled(mid INT, cid INT, enrollm...	0 row(s) affected	0.015 sec

AI Appendix

Prompt: In my ER diagram, there is a many-to-many relationship between Member and Class. Do I need to create a separate table (like Enrolled) for this relationship, or can I represent it directly within the existing tables?

AI Response: Yes, you need to create a separate table. In relational databases, many-to-many relationships cannot be represented directly. Therefore, a new relation (e.g., Enrolled) must be created, which includes the primary keys of both entities as foreign keys and typically uses them as a composite primary key.